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## SEMICONDUCTOR DEVICE AND SEMICONDUCTOR MEMORY DEVICE

### Abstract

A semiconductor device of embodiments includes: an n-channel MOSFET including a first gate insulating film and a first gate electrode, the first gate electrode including a first region containing Ti and Al and a second region provided between the first gate insulating film and the first region, in contact with the first gate insulating film, and containing a first metal element and nitrogen (N); and a p-channel MOSFET including a second gate insulating film and a second gate electrode, the second gate electrode including a third region containing Ti and Al, a fourth region provided between the second gate insulating film and the third region, in contact with the second gate insulating film, and containing the first metal element and nitrogen (N), and a fifth region provided between the third region and the fourth region and containing nitrogen (N) and a second metal element of Hf or Zr.

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## Background/Summary

### CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application is based upon and claims the benefit of priority from Japanese Patent Application No. 2024-032142, filed on Mar. 4, 2024, the entire contents of which are incorporated herein by reference.

### FIELD

[0002] Embodiments described herein relate generally to a semiconductor device and a semiconductor memory device.

### BACKGROUND

[0003] In a semiconductor device including an n-channel metal oxide semiconductor field effect transistor (MOSFET) and a p-channel MOSFET, it is desirable to control the threshold voltage of the n-channel MOSFET and the threshold voltage of the p-channel MOSFET to desired threshold voltages.

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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

[0004] FIG. 1 is a schematic cross-sectional view of a semiconductor device according to a first embodiment;

[0005] FIG. 2 is a diagram showing a method for manufacturing the semiconductor device according to the first embodiment;

[0006] FIG. 3 is a diagram showing a method for manufacturing the semiconductor device according to the first embodiment;

[0007] FIG. 4 is a diagram showing a method for manufacturing the semiconductor device according to the first embodiment;

[0008] FIG. 5 is a diagram showing a method for manufacturing the semiconductor device according to the first embodiment;

[0009] FIG. 6 is a diagram showing a method for manufacturing the semiconductor device according to the first embodiment;

[0010] FIG. 7 is a diagram showing a method for manufacturing the semiconductor device according to the first embodiment;

[0011] FIG. 8 is a schematic cross-sectional view of a semiconductor device according to a first comparative example;

[0012] FIG. 9 is a schematic cross-sectional view of a semiconductor device according to a second comparative example;

[0013] FIG. 10 is an explanatory diagram of the function and effect of the semiconductor device according to the first embodiment;

[0014] FIG. 11 is an explanatory diagram of the function and effect of the semiconductor device according to the first embodiment;

[0015] FIG. 12 is a schematic cross-sectional view of a semiconductor device according to a second embodiment;

[0016] FIG. 13 is a schematic cross-sectional view of a semiconductor device according to a third

embodiment; and

[0017] FIG. **14** is a schematic cross-sectional view of a semiconductor memory device according to a fourth embodiment.

#### DETAILED DESCRIPTION

[0018] A semiconductor device of embodiments includes: an n-channel MOSFET including a first gate insulating film and a first gate electrode, the first gate electrode including a first region and a second region, the second region provided between the first gate insulating film and the first region and in contact with the first gate insulating film, the first region containing titanium (Ti) and aluminum (Al), and the second region containing a first metal element and nitrogen (N); and a p-channel MOSFET including a second gate insulating film and a second gate electrode, the second gate electrode including a third region, a fourth region, and a fifth region, the fourth region provided between the second gate insulating film and the third region and in contact with the second gate insulating film, the fifth region provided between the third region and the fourth region, the third region containing titanium (Ti) and aluminum (Al), the fourth region containing the first metal element and nitrogen (N), and the fifth region containing nitrogen (N) and a second metal element being at least one of hafnium (Hf) and zirconium (Zr).

[0019] Hereinafter, embodiments will be described with reference to the diagrams. In addition, in the following description, the same or similar members and the like are denoted by the same reference numerals, and the description of the members and the like once described may be omitted as appropriate.

[0020] In addition, in this specification, the term “upper” or “lower” may be used for convenience. “Upper” or “lower” is a term indicating the relative positional relationship in the diagram, but is not a term that defines the positional relationship with respect to gravity.

[0021] The qualitative analysis and quantitative analysis of the chemical composition of members forming the semiconductor device and the semiconductor memory device in this specification can be performed by, for example, secondary ion mass spectrometry (SIMS), energy dispersive X-ray spectroscopy (EDX), and Rutherford back-scattering spectroscopy (RBS). In addition, when measuring the thickness of each member forming the semiconductor device and the semiconductor memory device, a distance between members, a crystal particle size, and the like, it is possible to use, for example, a transmission electron microscope (TEM).

#### First Embodiment

[0022] A semiconductor device according to a first embodiment includes an n-channel MOSFET including a first gate insulating film and a first gate electrode and a p-channel MOSFET including a second gate insulating film and a second gate electrode. The first gate electrode includes a first region containing titanium (Ti) and aluminum (Al) and a second region provided between the first gate insulating film and the first region, in contact with the first gate insulating film, and containing a first metal element and nitrogen (N). The second gate electrode includes a third region containing titanium (Ti) and aluminum (Al), a fourth region provided between the second gate insulating film and the third region, in contact with the second gate insulating film, and containing the first metal element and nitrogen (N), and a fifth region provided between the third region and the fourth region and containing nitrogen (N) and at least one second metal element of hafnium (Hf) and zirconium (Zr).

[0023] FIG. **1** is a schematic cross-sectional view of the semiconductor device according to the first embodiment.

[0024] The semiconductor device according to the first embodiment is a complementary metal oxide semiconductor (CMOS) semiconductor device **100**.

[0025] The CMOS semiconductor device **100** includes an n-channel MOSFET **100a** and a p-channel MOSFET **100b**. The n-channel MOSFET **100a** is a field effect transistor that uses electrons as carriers. The p-channel MOSFET **100b** is a field effect transistor that uses holes as carriers. In addition, the n-channel MOSFET **100a** and the p-channel MOSFET **100b** use a so-

called metal gate electrode in which a metal material is used for a gate electrode.

[0026] The CMOS semiconductor device **100** includes a CMOS circuit formed by, for example, the n-channel MOSFET **100a** and the p-channel MOSFET **100b**. The CMOS semiconductor device **100** includes, for example, a plurality of n-channel MOSFETs **100a** and a plurality of p-channel MOSFETs **100b**.

[0027] The n-channel MOSFET **100a** includes a silicon substrate **10**, a first gate insulating film **21**, a first gate electrode **31**, and an interlayer insulating layer **40**. The silicon substrate **10** includes a p-well region **10a** and a first source/drain region **10x**. The first gate electrode **31** includes a first lower region **31a**, a first intermediate region **31b**, a first upper region **31c**, and a first metal region **31d**.

[0028] The silicon substrate **10** is an example of a semiconductor substrate. The first lower region **31a** is an example of the second region. The first intermediate region **31b** is an example of the sixth region. The first upper region **31c** is an example of the first region.

[0029] The p-channel MOSFET **100b** includes the silicon substrate **10**, a second gate insulating film **22**, a second gate electrode **32**, and the interlayer insulating layer **40**. The silicon substrate **10** includes an n-well region **10b** and a second source/drain region **10y**. The second gate electrode **32** includes a second lower region **32a**, a second intermediate region **32b**, a second upper region **32c**, a second metal region **32d**, and a barrier region **32e**.

[0030] The second lower region **32a** is an example of the fourth region. The second intermediate region **32b** is an example of the seventh region. The second upper region **32c** is an example of the third region. The barrier region **32e** is an example of the fifth region.

[0031] The n-channel MOSFET **100a** includes the silicon substrate **10**, the first gate insulating film **21**, the first gate electrode **31**, and the interlayer insulating layer **40**. The n-channel MOSFET **100a** and the p-channel MOSFET **100b** are formed using the same silicon substrate **10**. The n-channel MOSFET **100a** includes a first portion of the silicon substrate **10**.

[0032] The p-well region **10a** is p-type silicon. In addition, the first source/drain region **10x** is n-type silicon. The p-well region **10a** and the first source/drain region **10x** are formed in the first portion of the silicon substrate **10**.

[0033] The first gate insulating film **21** is provided on the silicon substrate **10**. The first gate insulating film **21** is in contact with the silicon substrate **10**. The first gate insulating film **21** is provided between the silicon substrate **10** and the first gate electrode **31**. The first gate insulating film **21** faces the p-well region **10a** and the first source/drain region **10x**.

[0034] The first gate insulating film **21** is provided between the interlayer insulating layer **40** and the first gate electrode **31**. The first gate insulating film **21** is in contact with the interlayer insulating layer **40**.

[0035] The first gate insulating film **21** contains, for example, a material having a higher dielectric constant than silicon dioxide. The first gate insulating film **21** contains, for example, a High-k insulator. The first gate insulating film **21** is a so-called High-k gate insulating film.

[0036] The first gate insulating film **21** contains, for example, oxygen (**0**) and at least one element selected from a group consisting of hafnium (Hf), zirconium (Zr), tantalum (Ta), yttrium (Y), and titanium (Ti). The first gate insulating film **21** contains, for example, hafnium oxide, hafnium silicate, nitrogen incorporated hafnium silicate, nitrogen incorporated hafnium aluminate, zirconium oxide, tantalum oxide, yttrium oxide, or titanium oxide.

[0037] The film thickness of the first gate insulating film **21** is, for example, equal to or more than 1 nm and equal to or less than 10 nm. The equivalent oxide thickness of the first gate insulating film **21** is, for example, equal to or more than 0.5 nm and equal to or less than 2 nm.

[0038] The first gate insulating film **21** may have a stacked structure of a silicon oxide and a High-k insulator.

[0039] The first gate electrode **31** is provided on the first gate insulating film **21**. The first gate electrode **31** is in contact with the first gate insulating film **21**.

[0040] The first gate electrode **31** is formed by stacking, from the first gate insulating film **21**

upward, the first lower region **31a**, the first intermediate region **31b**, the first upper region **31c**, and the first metal region **31d** in this order. The first lower region **31a** is in contact with the first gate insulating film **21**. The first intermediate region **31b** is in contact with, for example, the first lower region **31a**. The first upper region **31c** is in contact with, for example, the first intermediate region **31b**. The first metal region **31d** is in contact with, for example, the first upper region **31c**.

[0041] The first lower region **31a** is provided between the first gate insulating film **21** and the first upper region **31c**. The first lower region **31a** is provided between the first gate insulating film **21** and the first intermediate region **31b**. The first lower region **31a** is provided on the first gate insulating film **21**. The first lower region **31a** is in contact with the first gate insulating film **21**.

[0042] The first lower region **31a** is electrically conductive. The first lower region **31a** contains a first metal element and nitrogen (N). The first metal element is, for example, a transition metal element. The first metal element is, for example, at least one metal element selected from a group consisting of titanium (Ti), tantalum (Ta), vanadium (V), chromium (Cr), zirconium (Zr), niobium (Nb), hafnium (Hf), and tungsten (W).

[0043] The first lower region **31a** contains, for example, titanium nitride, tantalum nitride, vanadium nitride, chromium nitride, zirconium nitride, niobium nitride, hafnium nitride, or tungsten nitride. The first lower region **31a** is, for example, a titanium nitride, a tantalum nitride, a vanadium nitride, a chromium nitride, a zirconium nitride, a niobium nitride, a hafnium nitride, or a tungsten nitride.

[0044] The thickness of the first lower region **31a** is, for example, equal to or more than 1 nm and equal to or less than 5 nm. In addition, in this specification, the thickness of each region is, for example, a thickness in a direction perpendicular to the surface of the silicon substrate **10**.

[0045] The first lower region **31a** is a region having a dominant effect on the threshold voltage of the n-channel MOSFET **100a**.

[0046] The first intermediate region **31b** is provided between the first upper region **31c** and the first lower region **31a**. The first intermediate region **31b** is provided on the first lower region **31a**. The first intermediate region **31b** is in contact with, for example, the first lower region **31a**.

[0047] The first intermediate region **31b** is electrically conductive. The first intermediate region **31b** contains a different material from the first lower region **31a**.

[0048] The first intermediate region **31b** contains a third metal element and nitrogen (N). The third metal element is different from the first metal element and the second metal element. The third metal element is, for example, a transition metal element. The third metal element is, for example, at least one metal element selected from a group consisting of titanium (Ti), tantalum (Ta), vanadium (V), chromium (Cr), zirconium (Zr), niobium (Nb), hafnium (Hf), and tungsten (W).

[0049] The first intermediate region **31b** contains, for example, titanium nitride, tantalum nitride, vanadium nitride, chromium nitride, zirconium nitride, niobium nitride, hafnium nitride, or tungsten nitride. The first intermediate region **31b** is, for example, a titanium nitride, a tantalum nitride, a vanadium nitride, a chromium nitride, a zirconium nitride, a niobium nitride, a hafnium nitride, or a tungsten nitride.

[0050] The thickness of the first intermediate region **31b** is, for example, equal to or more than 1 nm and equal to or less than 5 nm.

[0051] The first intermediate region **31b** functions as an etching stopper film when removing a film, which becomes the barrier region **32e** of the p-channel MOSFET **100b**, from the first gate electrode **31** during the manufacture of the n-channel MOSFET **100a**, for example.

[0052] The first upper region **31c** is provided between the first intermediate region **31b** and the first metal region **31d**. The first upper region **31c** is provided on the first intermediate region **31b**. The first upper region **31c** is in contact with, for example, the first intermediate region **31b**.

[0053] The first upper region **31c** is electrically conductive. The first upper region **31c** contains a different material from the first lower region **31a** and the first intermediate region **31b**.

[0054] The first upper region **31c** contains titanium (Ti) and aluminum (Al). The first upper region

**31c** contains, for example, a titanium-aluminum alloy.

[0055] The thickness of the first upper region **31c** is, for example, equal to or more than 1 nm and equal to or less than 8 nm.

[0056] The first upper region **31c** has a function of lowering the threshold voltage of the n-channel MOSFET **100a** when manufacturing the n-channel MOSFET **100a**, for example.

[0057] The first metal region **31d** is provided on the first upper region **31c**. The first metal region **31d** is in contact with, for example, the first upper region **31c**. The first upper region **31c** is provided between the first lower region **31a** and the first metal region **31d**.

[0058] The first metal region **31d** is electrically conductive. The electric resistivity of the first metal region **31d** is smaller than, for example, the electric resistivity of the first lower region **31a**, the electric resistivity of the first intermediate region **31b**, and the electric resistivity of the first upper region **31c**.

[0059] The first metal region **31d** contains a first metal. The first metal is, for example, at least one metal selected from a group consisting of aluminum (Al), tungsten (W), and molybdenum (Mo).

[0060] The first metal region **31d** may have, for example, a stacked structure of a barrier metal film and a metal film. The barrier metal film is, for example, a titanium nitride. The metal film is, for example, aluminum, tungsten, or molybdenum.

[0061] The first metal region **31d** has a function of reducing the gate resistance of the n-channel MOSFET **100a**, for example.

[0062] The p-channel MOSFET **100b** includes the silicon substrate **10**, the second gate insulating film **22**, the second gate electrode **32**, and the interlayer insulating layer **40**. The n-channel MOSFET **100a** and the p-channel MOSFET **100b** are formed using the same silicon substrate **10**. The p-channel MOSFET **100b** includes a second portion of the silicon substrate **10**.

[0063] The n-well region **10b** is n-type silicon. In addition, the second source/drain region **10y** is p-type silicon. The n-well region **10b** and the second source/drain region **10y** are formed in the second portion of the silicon substrate **10**.

[0064] The second gate insulating film **22** is provided on the silicon substrate **10**. The second gate insulating film **22** is in contact with the silicon substrate **10**. The second gate insulating film **22** is provided between the silicon substrate **10** and the second gate electrode **32**. The second gate insulating film **22** faces the n-well region **10b** and the second source/drain region **10y**.

[0065] The second gate insulating film **22** is provided between the interlayer insulating layer **40** and the second gate electrode **32**. The second gate insulating film **22** is in contact with the interlayer insulating layer **40**.

[0066] The second gate insulating film **22** contains the same material as the first gate insulating film **21**. The second gate insulating film **22** is formed of, for example, the same material as the first gate insulating film **21**.

[0067] The second gate insulating film **22** contains, for example, a material having a higher dielectric constant than silicon dioxide. The second gate insulating film **22** contains, for example, a High-k insulator. The second gate insulating film **22** is a so-called High-k gate insulating film.

[0068] The second gate insulating film **22** contains, for example, oxygen (O) and at least one element selected from a group consisting of hafnium (Hf), zirconium (Zr), tantalum (Ta), yttrium (Y), and titanium (Ti). The second gate insulating film **22** contains, for example, hafnium oxide, hafnium silicate, nitrogen incorporated hafnium silicate, nitrogen incorporated hafnium aluminate, zirconium oxide, tantalum oxide, yttrium oxide, or titanium oxide.

[0069] The thickness of the second gate insulating film **22** is, for example, equal to or more than 1 nm and equal to or less than 10 nm. The equivalent oxide thickness of the second gate insulating film **22** is, for example, equal to or more than 0.5 nm and equal to or less than 2 nm.

[0070] The second gate insulating film **22** may have a stacked structure of a silicon oxide and a High-k insulator.

[0071] The second gate electrode **32** is provided on the second gate insulating film **22**. The second

gate electrode **32** is in contact with the second gate insulating film **22**.

[0072] The second gate electrode **32** is formed by stacking, from the second gate insulating film **22** upward, the second lower region **32a**, the second intermediate region **32b**, the barrier region **32e**, the second upper region **32c**, and the second metal region **32d** in this order. The second lower region **32a** is in contact with the second gate insulating film **22**. The second intermediate region **32b** is in contact with, for example, the second lower region **32a**. The barrier region **32e** is in contact with, for example, the second intermediate region **32b**. The second upper region **32c** is in contact with, for example, the barrier region **32e**. The second metal region **32d** is in contact with, for example, the second upper region **32c**.

[0073] The stacked structure of the second gate electrode **32** is different from that of the first gate electrode **31** in that the stacked structure of the second gate electrode **32** includes the barrier region **32e**.

[0074] The second lower region **32a** is provided between the second gate insulating film **22** and the second upper region **32c**. The second lower region **32a** is provided between the second gate insulating film **22** and the second intermediate region **32b**. The second lower region **32a** is provided on the second gate insulating film **22**. The second lower region **32a** is in contact with the second gate insulating film **22**.

[0075] The second lower region **32a** contains the same material as the first lower region **31a**. The second lower region **32a** is formed of, for example, the same material as the first lower region **31a**.

[0076] The second lower region **32a** is electrically conductive. The second lower region **32a** contains a first metal element and nitrogen (N). The first metal element is, for example, a transition metal element. The first metal element is, for example, at least one metal element selected from a group consisting of titanium (Ti), tantalum (Ta), vanadium (V), chromium (Cr), zirconium (Zr), niobium (Nb), hafnium (Hf), and tungsten (W).

[0077] The second lower region **32a** contains, for example, titanium nitride, tantalum nitride, vanadium nitride, chromium nitride, zirconium nitride, niobium nitride, hafnium nitride, or tungsten nitride. The second lower region **32a** is, for example, a titanium nitride, a tantalum nitride, a vanadium nitride, a chromium nitride, a zirconium nitride, a niobium nitride, a hafnium nitride, or a tungsten nitride.

[0078] The thickness of the second lower region **32a** is, for example, equal to or more than 1 nm and equal to or less than 5 nm.

[0079] The second lower region **32a** is a region having a dominant effect on the threshold voltage of the p-channel MOSFET **100b**.

[0080] The second intermediate region **32b** is provided between the second upper region **32c** and the second lower region **32a**. The second intermediate region **32b** is provided on the second lower region **32a**. The second intermediate region **32b** is in contact with, for example, the second lower region **32a**. The second intermediate region **32b** is provided between the second lower region **32a** and the barrier region **32e**.

[0081] The second intermediate region **32b** is electrically conductive. The second intermediate region **32b** contains a different material from the second lower region **32a**.

[0082] The second intermediate region **32b** contains the same material as the first intermediate region **31b**. The second intermediate region **32b** is formed of, for example, the same material as the first intermediate region **31b**.

[0083] The second intermediate region **32b** contains a third metal element and nitrogen (N). The third metal element is different from the first metal element and the second metal element. The third metal element is, for example, a transition metal element. The third metal element is, for example, at least one metal element selected from a group consisting of titanium (Ti), tantalum (Ta), vanadium (V), chromium (Cr), zirconium (Zr), niobium (Nb), hafnium (Hf), and tungsten (W).

[0084] The second intermediate region **32b** contains, for example, titanium nitride, tantalum

nitride, vanadium nitride, chromium nitride, zirconium nitride, niobium nitride, hafnium nitride, or tungsten nitride. The second intermediate region **32b** is, for example, a titanium nitride, a tantalum nitride, a vanadium nitride, a chromium nitride, a zirconium nitride, a niobium nitride, a hafnium nitride, or a tungsten nitride.

[0085] The thickness of the second intermediate region **32b** is, for example, equal to or more than 1 nm and equal to or less than 5 nm.

[0086] The barrier region **32e** is provided between the second upper region **32c** and the second lower region **32a**. The barrier region **32e** is provided between the second upper region **32c** and the second intermediate region **32b**. The barrier region **32e** is provided on the second intermediate region **32b**. The barrier region **32e** is in contact with, for example, the second intermediate region **32b**.

[0087] The barrier region **32e** is electrically conductive. The barrier region **32e** contains, for example, a different material from the second lower region **32a**, the second intermediate region **32b**, and the second upper region **32c**.

[0088] The barrier region **32e** contains nitrogen (N) and at least one second metal element of hafnium (Hf) and zirconium (Zr). The second metal element is different from, for example, the third metal element contained in the second intermediate region **32b**.

[0089] The barrier region **32e** contains, for example, hafnium nitride or zirconium nitride. The barrier region **32e** is, for example, a hafnium nitride or a zirconium nitride.

[0090] The atomic concentration of nitrogen (N) contained in the barrier region **32e** is, for example, equal to or more than 1 time and equal to or less than 1.3 times the atomic concentration of at least one of the second metal elements contained in the barrier region **32e**. The atomic concentration of nitrogen (N) contained in the barrier region **32e** is, for example, equal to or more than 1.1 times and equal to or less than 1.3 times the atomic concentration of at least one of the second metal elements contained in the barrier region **32e**.

[0091] The thickness of the barrier region **32e** is, for example, equal to or more than 1 nm and equal to or less than 10 nm. The thickness of the barrier region **32e** is larger than, for example, the thickness of the second lower region **32a** and the thickness of the second intermediate region **32b**.

[0092] The barrier region **32e** has, for example, a function of suppressing an increase in the threshold voltage of the p-channel MOSFET **100b** by suppressing the movement of nitrogen atoms or aluminum atoms in the second gate electrode **32** during the manufacture of the p-channel MOSFET **100b**.

[0093] The second upper region **32c** is provided between the barrier region **32e** and the second metal region **32d**. The second upper region **32c** is provided on the barrier region **32e**. The second upper region **32c** is in contact with, for example, the barrier region **32e**.

[0094] The second upper region **32c** is electrically

[0095] conductive. The second upper region **32c** contains a different material from the second lower region **32a** and the second intermediate region **32b**.

[0096] The second upper region **32c** contains the same material as the first upper region **31c**. The second upper region **32c** is formed of, for example, the same material as the first upper region **31c**.

[0097] The second upper region **32c** contains titanium (Ti) and aluminum (Al). The second upper region **32c** contains, for example, a titanium-aluminum alloy.

[0098] The thickness of the second upper region **32c** is, for example, equal to or more than 1 nm and equal to or less than 8 nm.

[0099] The second metal region **32d** is provided on the second upper region **32c**. The second metal region **32d** is in contact with, for example, the second upper region **32c**. The second upper region **32c** is provided between the barrier region **32e** and the second metal region **32d**.

[0100] The second metal region **32d** is electrically conductive. The electric resistivity of the second metal region **32d** is smaller than, for example, the electric resistivity of the second lower region **32a**, the electric resistivity of the second intermediate region **32b**, and the electric resistivity of the

second upper region **32c**.

[0101] The second metal region **32d** contains the same material as the first metal region **31d**. The second metal region **32d** is formed of, for example, the same material as the first metal region **31d**.

[0102] The second metal region **32d** contains a first metal. The first metal is, for example, at least one metal selected from a group consisting of aluminum (Al), tungsten (W), and molybdenum (Mo).

[0103] The second metal region **32d** may have, for example, a stacked structure of a barrier metal film and a metal film. The barrier metal film is, for example, a titanium nitride. The metal film is, for example, aluminum, tungsten, or molybdenum.

[0104] The second metal region **32d** has a function of reducing the gate resistance of the p-channel MOSFET **100b**, for example.

[0105] Next, an example of a method for manufacturing the semiconductor device according to the first embodiment will be described. An example of the method for manufacturing the CMOS semiconductor device **100** according to the first embodiment is a manufacturing method known as a so-called gate-last method. In the gate-last manufacturing method, a source/drain region is formed after a dummy gate electrode is formed. Then, after the source/drain region is formed, the dummy gate electrode is removed and replaced with a gate electrode.

[0106] FIGS. **2** to **7** are diagrams showing a method for manufacturing the semiconductor device according to the first embodiment. FIGS. **2** to **7** show cross sections corresponding to FIG. **1**.

[0107] Hereinafter, an example will be described in which the first gate insulating film **21** and the second gate insulating film **22** are formed of hafnium oxide, the first lower region **31a** and the second lower region **32a** are formed of titanium nitride, the first intermediate region **31b** and the second intermediate region **32b** are formed of tantalum nitride, the first upper region **31c** and the second upper region **32c** are formed of a titanium-aluminum alloy, the first metal region **31d** and the second metal region **32d** are a stacked structure of titanium nitride and aluminum, and the barrier region **32e** is formed of hafnium nitride.

[0108] First, the silicon substrate **10** is prepared. Then, the p-well region **10a** and the n-well region **10b** are formed in the silicon substrate **10**. The p-well region **10a** and the n-well region **10b** are formed by using, for example, a lithography method, an ion implantation method, and a heat treatment.

[0109] Then, a dummy gate electrode **50** is formed on the silicon substrate **10**. The dummy gate electrode **50** is formed of, for example, polycrystalline silicon. The dummy gate electrode **50** is formed by using, for example, a chemical vapor deposition (CVD) method, a lithography method, and a reactive ion etching (RIE) method. Then, ion implantation is performed using the dummy gate electrode **50** as a mask. Arsenic (As) ions are implanted into the p-well region **10a**, and boron (B) ions are implanted into the n-well region **10b**. Thereafter, a heat treatment is performed to activate arsenic (As) and boron (B), thereby forming the first source/drain region **10x** of n-type and the second source/drain region **10y** of p-type (FIG. **2**).

[0111] Then, the interlayer insulating layer **40** is formed on the side surfaces of the dummy gate electrode **50** (FIG. **3**). The interlayer insulating layer **40** is, for example, a silicon oxide. The interlayer insulating layer **40** is formed by using, for example, a CVD method and a chemical mechanical polishing (CMP) method.

[0112] Then, the dummy gate electrode **50** is removed to form a groove **51** in the interlayer insulating layer **40** (FIG. **4**). The dummy gate electrode **50** is removed by using, for example, a wet etching method.

[0113] Then, a hafnium oxide film **52** is formed in the groove **51**. The hafnium oxide film **52** is formed by using, for example, an atomic layer deposition (ALD) method. The hafnium oxide film **52** eventually becomes the first gate insulating film **21** and the second gate insulating film **22**.

[0114] In addition, before forming the hafnium oxide film **52**, a silicon oxide film (not shown) may be formed on the surface of the silicon substrate **10**. The silicon oxide film is formed, for example,

by thermally oxidizing the surface of the silicon substrate **10**.

[0115] Then, a titanium nitride film **53**, a tantalum nitride film **54**, and a hafnium nitride film **55** are formed on the hafnium oxide film **52** (FIG. 5). The titanium nitride film **53**, the tantalum nitride film **54**, and the hafnium nitride film **55** are formed by using, for example, the ALD method. The titanium nitride film **53** eventually becomes the first lower region **31a** and the second lower region **32a**. The tantalum nitride film **54** eventually becomes the first intermediate region **31b** and the second intermediate region **32b**. The hafnium nitride film **55** eventually becomes the barrier region **32e**.

[0116] Then, the hafnium nitride film **55** formed in the groove **51** on the p-well region **10a** is removed. The hafnium nitride film **55** formed in the groove **51** on the n-well region **10b** remains (FIG. 6). The hafnium nitride film **55** is removed by using, for example, a lithography method and a wet etching method.

[0117] When removing the hafnium nitride film **55** formed in the groove **51** on the p-well region **10a**, the tantalum nitride film **54** functions as an etching stopper film.

[0118] Then, a titanium-aluminum alloy film **56** is formed on the tantalum nitride film **54** and the hafnium nitride film **55**. In addition, a stacked film **57** of titanium nitride and aluminum is formed on the titanium-aluminum alloy film **56** (FIG. 7). The titanium-aluminum alloy film **56** is formed by using, for example, the ALD method. The stacked film **57** of titanium nitride and aluminum is formed by using, for example, the CVD method.

[0119] Then, the hafnium oxide film **52**, the titanium nitride film **53**, the tantalum nitride film **54**, the hafnium nitride film **55**, the titanium-aluminum alloy film **56**, and the stacked film **57** on the interlayer insulating layer **40** are removed. The hafnium oxide film **52**, the titanium nitride film **53**, the tantalum nitride film **54**, the hafnium nitride film **55**, the titanium-aluminum alloy film **56**, and the stacked film **57** are removed by using, for example, the chemical mechanical polishing (CMP) method.

[0120] By the manufacturing method described above, the CMOS semiconductor device **100** shown in FIG. 1 is manufactured.

[0121] Next, the function and effect of the semiconductor device according to the first embodiment will be described.

[0122] In a semiconductor device including an n-channel MOSFET and a p-channel MOSFET, it is desirable to control the threshold voltage of the n-channel MOSFET and the threshold voltage of the p-channel MOSFET to desired values.

[0123] FIG. 8 is a schematic cross-sectional view of a semiconductor device according to a first comparative example. The semiconductor device according to the first comparative example is a CMOS semiconductor device **901**. The CMOS semiconductor device **901** includes an n-channel MOSFET **901a** and a p-channel MOSFET **901b**.

[0124] The CMOS semiconductor device **901** is different from the CMOS semiconductor device **100** according to the first embodiment in that the first gate electrode **31** of the n-channel MOSFET **901a** includes only the first lower region **31a** and the first metal region **31d** and the second gate electrode **32** of the p-channel MOSFET **901b** includes only the second lower region **32a** and the second metal region **32d**.

[0125] In the CMOS semiconductor device **901** according to the first comparative example, for example, a case will be considered in which the first gate insulating film **21** and the second gate insulating film **22** are hafnium oxide films and the first lower region **31a** and the second lower region **32a** are titanium nitride regions. In this case, the threshold voltage of the n-channel MOSFET **901a** is higher than a desired threshold voltage  $V_{thn}$ . On the other hand, the threshold voltage of the p-channel MOSFET **901b** is a desired threshold voltage  $V_{thp}$ .  $V_{thn}$  is, for example, in the range of 0.2 V to 0.4 V, and  $V_{thp}$  is, for example, in the range of -0.2 V to -0.4 V.

[0126] In the CMOS semiconductor device **901**, it is thought that the work function of the titanium nitride forming the first lower region **31a** and the second lower region **32a** has a dominant effect on

the threshold voltage of the n-channel MOSFET **901a** and the threshold voltage of the p-channel MOSFET **901b**.

[0127] In the following description, a high threshold voltage means that the absolute value of the threshold voltage is large. In addition, a low threshold voltage means that the absolute value of the threshold voltage is small.

[0128] FIG. **9** is a schematic cross-sectional view of a semiconductor device according to a second comparative example. The semiconductor device according to the second comparative example is a CMOS semiconductor device **902**. The CMOS semiconductor device **902** includes an n-channel MOSFET **902a** and a p-channel MOSFET **902b**.

[0129] The CMOS semiconductor device **902** is different from the CMOS semiconductor device **901** according to the first comparative example in that the first gate electrode **31** of the n-channel MOSFET **902a** includes the first intermediate region **31b** and the first upper region **31c**. In addition, the CMOS semiconductor device **902** is different from the CMOS semiconductor device **901** according to the first comparative example in that the second gate electrode **32** of the p-channel MOSFET **902b** includes the second intermediate region **32b** and the second upper region **32c**.

[0130] The CMOS semiconductor device **902** is different from the CMOS semiconductor device **100** according to the first embodiment in that the second gate electrode **32** of the p-channel MOSFET **100b** does not include the barrier region **32e**.

[0131] In the CMOS semiconductor device **902**, a case will be considered in which the first gate insulating film **21** and the second gate insulating film **22** are hafnium oxide films, the first lower region **31a** and the second lower region **32a** are titanium nitride regions, the first intermediate region **31b** and the second intermediate region **32b** are tantalum nitride regions, and the first upper region **31c** and the second upper region **32c** are titanium-aluminum alloy regions. In the CMOS semiconductor device **902**, the first gate electrode **31** and the second gate electrode **32** have the same stacked structure.

[0132] In the CMOS semiconductor device **902** according to the second comparative example, the threshold voltage of the n-channel MOSFET **902a** is the desired threshold voltage  $V_{thn}$ . On the other hand, the threshold voltage of the p-channel MOSFET **902b** is higher than the desired threshold voltage  $V_{thp}$ . In other words, in the CMOS semiconductor device **902** according to the second comparative example, the threshold voltage of the n-channel MOSFET **902a** is lower than the threshold voltage of the n-channel MOSFET **901a** of the CMOS semiconductor device **901** according to the first comparative example. On the other hand, the threshold voltage of the p-channel MOSFET **902b** is higher than the threshold voltage  $V_{thp}$  of the p-channel MOSFET **901b** of the CMOS semiconductor device **901** according to the first comparative example.

[0133] The reason why, in the CMOS semiconductor device **902**, the threshold voltage of the n-channel MOSFET is lower and the threshold voltage of the p-channel MOSFET is higher than in the CMOS semiconductor device **901** is thought that the effective work function of the gate electrode changes due to the movement of atoms in the gate electrode.

[0134] The effective work function is different from the work function, which is a value specific to a material. The work function is an energy difference between the vacuum level and the Fermi level. The effective work function means a work function obtained by actually electrically measuring the MOS structure. For example, the effective work function is a value that can be changed by changing the stacked structure of the gate electrode.

[0135] FIG. **10** is an explanatory diagram of the function and effect of the semiconductor device according to the first embodiment. FIG. **10** is an enlarged schematic cross-sectional view including a part (left) of the first gate electrode **31** of the n-channel MOSFET **902a** and a part (right) of the second gate electrode **32** of the p-channel MOSFET **902b** in the CMOS semiconductor device **902** according to the second comparative example.

[0136] FIG. **10** schematically shows the movement of nitrogen atoms and aluminum atoms in a

gate electrode. As shown in FIG. 10, it is thought that nitrogen atoms or aluminum atoms move through the gate electrode due to the heat treatment applied during the manufacture of the CMOS semiconductor device 902.

[0137] For example, nitrogen atoms contained in the titanium nitride region move to the titanium-aluminum alloy region, decreasing the nitrogen concentration in the titanium nitride region. In addition, for example, aluminum atoms in the titanium-aluminum alloy region move to the titanium nitride region, increasing the aluminum concentration in the titanium nitride region. The titanium-aluminum alloy region functions as a nitrogen absorber. In addition, the titanium-aluminum alloy region functions as an aluminum emission source.

[0138] It is thought that the effective work function of the gate electrode decreases when the nitrogen concentration in the titanium nitride region decreases or the aluminum concentration in the titanium nitride region increases. It is thought that, as the effective work function of the gate electrode decreases, the threshold voltage of the n-channel MOSFET 902a decreases and the threshold voltage of the p-channel MOSFET 902b increases.

[0139] In the CMOS semiconductor device 100 according to the first embodiment, the second gate electrode 32 of the p-channel MOSFET 100b includes the barrier region 32e. In the CMOS semiconductor device 100 according to the first embodiment, the threshold voltage of the n-channel MOSFET 100a is the desired threshold voltage  $V_{thn}$ . In addition, the threshold voltage of the p-channel MOSFET 100b is also the desired threshold voltage  $V_{thp}$ .

[0140] FIG. 11 is an explanatory diagram of the function and effect of the semiconductor device according to the first embodiment. FIG. 11 is an enlarged schematic cross-sectional view including a part (left) of the first gate electrode 31 of the n-channel MOSFET 100a and a part (right) of the second gate electrode 32 of the p-channel MOSFET 100b in the CMOS semiconductor device 100 according to the first embodiment. FIG. 11 is a diagram corresponding to FIG. 10.

[0141] In the CMOS semiconductor device 100, the second gate electrode 32 of the p-channel MOSFET 100b includes a hafnium nitride region that is the barrier region 32e. As shown in FIG. 11, it is thought that the movement of nitrogen atoms and aluminum atoms in the gate electrode is suppressed by providing the hafnium nitride region. Therefore, it is thought that a decrease in nitrogen concentration in the titanium nitride region or an increase in aluminum concentration in the titanium nitride region as in the gate electrode of the n-channel MOSFET 100a is suppressed. Therefore, it is thought that the threshold voltage of the p-channel MOSFET 100b is the desired threshold voltage  $V_{thp}$ , like the p-channel MOSFET 901b in the first comparative example.

[0142] The reason why the movement of nitrogen atoms in the gate electrode is suppressed when the barrier region 32e contains hafnium nitride is thought that nitrogen atoms are unlikely to escape because the generation energy for nitrogen vacancies in hafnium nitride is high. For example, hafnium nitride has a higher nitrogen vacancy generation energy than titanium nitride.

[0143] In addition, the reason why the movement of aluminum atoms in the gate electrode is suppressed when the barrier region 32e contains hafnium nitride is thought that aluminum is unlikely to exist in hafnium nitride because the generation energy required to substitute hafnium in hafnium nitride with aluminum is high. For example, hafnium nitride has a higher generation energy for aluminum substitution of metal elements than titanium nitride.

[0144] Similarly to hafnium nitride, zirconium nitride has a higher nitrogen vacancy generation energy than titanium nitride. In addition, similarly to hafnium nitride, zirconium nitride has a higher generation energy for aluminum substitution of metal elements than titanium nitride. Therefore, it is thought that the movement of nitrogen atoms and aluminum atoms in the gate electrode is suppressed because the barrier region 32e contains zirconium nitride.

[0145] In the CMOS semiconductor device 100 according to the first embodiment, it is possible to control the threshold voltage of the n-channel MOSFET 100a and the threshold voltage of the p-channel MOSFET 100b to desired threshold voltages.

[0146] From the viewpoint of suppressing an increase in the threshold voltage of the p-channel

MOSFET **100b**, the thickness of the barrier region **32e** is preferably equal to or more than 1 nm, more preferably equal to or more than 3 nm, and even more preferably equal to or more than 5 nm. [0147] From the viewpoint of suppressing an increase in the threshold voltage of the p-channel MOSFET **100b**, the thickness of the barrier region **32e** is preferably larger than the thickness of the second lower region **32a** and the thickness of the second intermediate region **32b**.

[0148] From the viewpoint of suppressing an increase in the threshold voltage of the p-channel MOSFET **100b**, the atomic concentration of nitrogen (N) contained in the barrier region **32e** is preferably equal to or more than 1 time, more preferably equal to or more than 1.1 times, and even more preferably equal to or more than 1.2 times the atomic concentration of the second metal element contained in the barrier region **32e**.

[0149] From the viewpoint of lowering the threshold voltage of the n-channel MOSFET **100a**, the thickness of the first upper region **31c** is preferably equal to or more than 1 nm, more preferably equal to or more than 3 nm, and even more preferably equal to or more than 5 nm.

[0150] As described above, according to the semiconductor device according to the first embodiment, it is possible to provide a semiconductor device in which the threshold voltage of the n-channel MOSFET and the threshold voltage of the p-channel MOSFET are controlled to desired threshold voltages.

## Second Embodiment

[0151] A semiconductor device according to a second embodiment is different from the semiconductor device according to the first embodiment in that the first gate electrode does not include the sixth region and the second gate electrode does not include the seventh region. Hereinafter, the description of a part of the content overlapping the first embodiment will be omitted.

[0152] FIG. **12** is a schematic cross-sectional view of the semiconductor device according to the second embodiment. The semiconductor device according to the second embodiment is a CMOS semiconductor device **200**.

[0153] The CMOS semiconductor device **200** includes an n-channel MOSFET **200a** and a p-channel MOSFET **200b**. The n-channel MOSFET **200a** is a field effect transistor that uses electrons as carriers. The p-channel MOSFET **200b** is a field effect transistor that uses holes as carriers. In addition, the n-channel MOSFET **200a** and the p-channel MOSFET **200b** use a so-called metal gate electrode in which a metal material is used for a gate electrode.

[0154] The CMOS semiconductor device **200** includes a CMOS circuit formed by the n-channel MOSFET **200a** and the p-channel MOSFET **200b**. The CMOS semiconductor device **200** includes, for example, a plurality of n-channel MOSFETs **200a** and a plurality of p-channel MOSFETs **200b**.

[0155] The n-channel MOSFET **200a** includes a silicon substrate **10**, a first gate insulating film **21**, a first gate electrode **31**, and an interlayer insulating layer **40**. The silicon substrate **10** includes a p-well region **10a** and a first source/drain region **10x**. The first gate electrode **31** includes a first lower region **31a**, a first upper region **31c**, and a first metal region **31d**.

[0156] The silicon substrate **10** is an example of a semiconductor substrate. The first lower region **31a** is an example of the second region. The first upper region **31c** is an example of the first region.

[0157] The p-channel MOSFET **200b** includes the silicon substrate **10**, a second gate insulating film **22**, a second gate electrode **32**, and the interlayer insulating layer **40**. The silicon substrate **10** includes an n-well region **10b** and a second source/drain region **10y**. The second gate electrode **32** includes a second lower region **32a**, a second upper region **32c**, a second metal region **32d**, and a barrier region **32e**.

[0158] The second lower region **32a** is an example of the fourth region. The second upper region **32c** is an example of the third region. The barrier region **32e** is an example of the fifth region.

[0159] The first upper region **31c** is provided on the first lower region **31a**. The first upper region **31c** is in contact with the first lower region **31a**.

[0160] The barrier region **32e** is provided on the second lower region **32a**. The barrier region **32e** is

in contact with the second lower region **32a**.

[0161] The first lower region **31a** and the second lower region **32a** contain a first metal element and nitrogen (N). In addition, the barrier region **32e** contains nitrogen (N) and at least one second metal element of hafnium (Hf) and zirconium (Zr).

[0162] In the CMOS semiconductor device **100** according to the first embodiment, the first intermediate region **31b** functions as an etching stopper film when removing a film, which becomes the barrier region **32e** of the p-channel MOSFET **100b**, from the first gate electrode **31** during the manufacture of the n-channel MOSFET **100a**. When the first lower region **31a** functions as an etching stopper film when removing a film to become the barrier region **32e** of the second gate electrode **32** from the first gate electrode **31**, it is possible to omit the first intermediate region **31b** and the second intermediate region **32b** of the CMOS semiconductor device **100**, as in the CMOS semiconductor device **200** according to the second embodiment.

[0163] As described above, according to the semiconductor device according to the second embodiment, as in the semiconductor device according to the first embodiment, it is possible to provide a semiconductor device in which the threshold voltage of the n-channel MOSFET and the threshold voltage of the p-channel MOSFET are controlled to desired threshold voltages.

### Third Embodiment

[0164] A semiconductor device according to a third embodiment is different from the semiconductor device according to the first embodiment in that the semiconductor device according to the third embodiment is manufactured using a manufacturing method known as a so-called gate-first method. Hereinafter, the description of a part of the content overlapping the first embodiment will be omitted.

[0165] FIG. **13** is a schematic cross-sectional view of the semiconductor device according to the third embodiment. The semiconductor device according to the third embodiment is a CMOS semiconductor device **300**.

[0166] The CMOS semiconductor device **300** includes an n-channel MOSFET **300a** and a p-channel MOSFET **300b**. The n-channel MOSFET **300a** is a field effect transistor that uses electrons as carriers. The p-channel MOSFET **300b** is a field effect transistor that uses holes as carriers. In addition, the n-channel MOSFET **300a** and the p-channel MOSFET **300b** use a so-called metal gate electrode in which a metal material is used for a gate electrode.

[0167] The CMOS semiconductor device **300** includes a CMOS circuit formed by the n-channel MOSFET **300a** and the p-channel MOSFET **300b**. The CMOS semiconductor device **300** includes, for example, a plurality of n-channel MOSFETs **300a** and a plurality of p-channel MOSFETs **300b**.

[0168] The n-channel MOSFET **300a** includes a silicon substrate **10**, a first gate insulating film **21**, and a first gate electrode **31**. The silicon substrate **10** includes a p-well region **10a** and a first source/drain region **10x**. The first gate electrode **31** includes a first lower region **31a**, a first intermediate region **31b**, a first upper region **31c**, and a first metal region **31d**.

[0169] The silicon substrate **10** is an example of a semiconductor substrate. The first lower region **31a** is an example of the second region. The first intermediate region **31b** is an example of the sixth region. The first upper region **31c** is an example of the first region.

[0170] The p-channel MOSFET **300b** includes the silicon substrate **10**, a second gate insulating film **22**, and a second gate electrode **32**. The silicon substrate **10** includes an n-well region **10b** and a second source/drain region **10y**. The second gate electrode **32** includes a second lower region **32a**, a second intermediate region **32b**, a second upper region **32c**, a second metal region **32d**, and a barrier region **32e**.

[0171] The second lower region **32a** is an example of the fourth region. The second intermediate region **32b** is an example of the seventh region. The second upper region **32c** is an example of the third region. The barrier region **32e** is an example of the fifth region.

[0172] The CMOS semiconductor device **300** is manufactured using a manufacturing method

known as a so-called gate-first method. In the gate-first manufacturing method, the source/drain region is formed after the gate electrode is formed. For example, after the gate electrode is formed using a lithography method and an RIE method, ion implantation is performed using the gate electrode as a mask to form the source/drain region.

[0173] As described above, according to the semiconductor device according to the third embodiment, as in the semiconductor device according to the first embodiment, it is possible to provide a semiconductor device in which the threshold voltage of the n-channel MOSFET and the threshold voltage of the p-channel MOSFET are controlled to desired threshold voltages.

#### Fourth Embodiment

[0174] A semiconductor memory device according to a fourth embodiment includes an n-channel MOSFET including a first gate insulating film and a first gate electrode, a p-channel MOSFET including a second gate insulating film and a second gate electrode, and a memory cell array including a plurality of memory cells. The first gate electrode includes a first region containing titanium (Ti) and aluminum (Al) and a second region provided between the first gate insulating film and the first region, in contact with the first gate insulating film, and containing a first metal element and nitrogen (N). The second gate electrode includes a third region containing titanium (Ti) and aluminum (Al), a fourth region provided between the second gate insulating film and the third region, in contact with the second gate insulating film, and containing the first metal element and nitrogen (N), and a fifth region provided between the third region and the fourth region and containing nitrogen (N) and at least one second metal element of hafnium (Hf) and zirconium (Zr). The semiconductor memory device according to the fourth embodiment is different from the first embodiment in that, in addition to the semiconductor device according to the first embodiment, a memory cell array including a plurality of memory cells is provided. Hereinafter, the description of a part of the content overlapping the first embodiment will be omitted.

[0175] FIG. **14** is a schematic cross-sectional view of the semiconductor memory device according to the fourth embodiment. The semiconductor memory device according to the fourth embodiment is a three-dimensional NAND flash memory **400**.

[0176] The three-dimensional NAND flash memory **400** includes the CMOS semiconductor device **100** according to the first embodiment and a memory layer **80**. The memory layer **80** includes a memory cell array **80x** including a plurality of memory cells MC. The plurality of memory cells MC are arranged, for example, three-dimensionally in the memory cell array **80x**.

[0177] The CMOS semiconductor device **100** and the memory layer **80** are electrically connected to each other using, for example, metal wiring. The CMOS semiconductor device **100** and the memory cell array **80x** are electrically connected to each other using, for example, metal wiring.

[0178] The n-channel MOSFET **100a** and the p-channel MOSFET **100b** included in the CMOS semiconductor device **100** control the operations of the plurality of memory cells MC included in the memory cell array **80x**.

[0179] The three-dimensional NAND flash memory **400** can be manufactured, for example, by bonding the CMOS semiconductor device **100** and the memory layer **80** manufactured separately to each other.

[0180] As described above, according to the semiconductor memory device according to the fourth embodiment, it is possible to provide a semiconductor memory device in which the threshold voltage of the n-channel MOSFET and the threshold voltage of the p-channel MOSFET are controlled to desired threshold voltages.

[0181] In the first to third embodiments, the silicon substrate has been described as an example of the semiconductor substrate, but it is also possible to use a semiconductor substrate other than the silicon substrate. In the first to third embodiments, the case where

[0182] the n-channel MOSFET **100a** and the p-channel MOSFET **100b** are formed using the same semiconductor substrate has been described as an example. However, for example, the n-channel MOSFET **100a** and the p-channel MOSFET **100b** can also be formed using semiconductor layers

that are physically separated from each other.

[0183] In the fourth embodiment, a semiconductor memory device including the CMOS semiconductor device **100** according to the first embodiment has been described as an example. However, the semiconductor memory device according to the fourth embodiment may be a semiconductor memory device including the CMOS semiconductor device according to the second or third embodiment.

[0184] In the fourth embodiment, a three-dimensional NAND flash memory has been described as an example, but the semiconductor memory device may be, for example, other semiconductor memory devices such as a dynamic random access memory (DRAM) and a resistance change memory.

[0185] While certain embodiments have been described, these embodiments have been presented by way of example only, and are not intended to limit the scope of the inventions. Indeed, the semiconductor device and the semiconductor memory device described herein may be embodied in a variety of other forms; furthermore, various omissions, substitutions and changes in the form of the devices and methods described herein may be made without departing from the spirit of the inventions. The accompanying claims and their equivalents are intended to cover such forms or modifications as would fall within the scope and spirit of the inventions.

## Claims

1. A semiconductor device, comprising: an n-channel MOSFET including a first gate insulating film and a first gate electrode, the first gate electrode including a first region and a second region, the second region provided between the first gate insulating film and the first region and in contact with the first gate insulating film, the first region containing titanium (Ti) and aluminum (Al), and the second region containing a first metal element and nitrogen (N); and a p-channel MOSFET including a second gate insulating film and a second gate electrode, the second gate electrode including a third region, a fourth region, and a fifth region, the fourth region provided between the second gate insulating film and the third region and in contact with the second gate insulating film, the fifth region provided between the third region and the fourth region, the third region containing titanium (Ti) and aluminum (Al), the fourth region containing the first metal element and nitrogen (N), and the fifth region containing nitrogen (N) and a second metal element being at least one of hafnium (Hf) and zirconium (Zr).
2. The semiconductor device according to claim 1, wherein the first metal element and the second metal element are different.
3. The semiconductor device according to claim 1, wherein the first gate electrode further includes a sixth region provided between the first region and the second region and containing a third metal element and nitrogen (N), the third metal element being different from the first metal element and the second metal element, and the second gate electrode further includes a seventh region provided between the fourth region and the fifth region and containing the third metal element and nitrogen (N).
4. The semiconductor device according to claim 1, wherein the first metal element is a transition metal element.
5. The semiconductor device according to claim 4, wherein the first metal element is at least one metal element selected from a group consisting of titanium (Ti), tantalum (Ta), vanadium (V), chromium (Cr), zirconium (Zr), niobium (Nb), hafnium (Hf), and tungsten (W).
6. The semiconductor device according to claim 3, wherein the third metal element is a transition metal element.
7. The semiconductor device according to claim 6, wherein the third metal element is at least one metal element selected from a group consisting of titanium (Ti), tantalum (Ta), vanadium (V), chromium (Cr), zirconium (Zr), niobium (Nb), hafnium (Hf), and tungsten (W).

8. The semiconductor device according to claim 1, wherein the first gate insulating film and the second gate insulating film contain a material having a higher dielectric constant than silicon dioxide.
9. The semiconductor device according to claim 1, wherein the first gate insulating film and the second gate insulating film contain oxygen (O) and at least one element selected from a group consisting of hafnium (Hf), zirconium (Zr), tantalum (Ta), yttrium (Y), and titanium (Ti).
10. The semiconductor device according to claim 1, wherein the n-channel MOSFET further includes a first portion of a semiconductor substrate, and the first gate insulating film is provided on the first portion, and the p-channel MOSFET further includes a second portion of the semiconductor substrate, and the second gate insulating film is provided on the second portion.
11. The semiconductor device according to claim 1, wherein the first gate electrode further includes a first metal region containing a first metal, the second gate electrode further includes a second metal region containing the first metal, the first region is provided between the second region and the first metal region, the third region is provided between the fifth region and the second metal region, and the first metal is at least one metal selected from a group consisting of aluminum (Al), tungsten (W), and molybdenum (Mo).
12. The semiconductor device according to claim 1, wherein an atomic concentration of nitrogen (N) contained in the fifth region is equal to or more than 1 time and equal to or less than 1.3 times an atomic concentration of the second metal element contained in the fifth region.
13. The semiconductor device according to claim 3, wherein a thickness of the fifth region is larger than a thickness of the fourth region and larger than a thickness of the seventh region.
14. The semiconductor device according to claim 3, wherein the first metal element is titanium (Ti), and the third metal element is tantalum (Ta).
15. A semiconductor memory device, comprising: an n-channel MOSFET including a first gate insulating film and a first gate electrode, the first gate electrode including a first region and a second region, the second region provided between the first gate insulating film and the first region and in contact with the first gate insulating film, the first region containing titanium (Ti) and aluminum (Al), and the second region containing a first metal element and nitrogen (N); a p-channel MOSFET including a second gate insulating film and a second gate electrode, the second gate electrode including a third region, a fourth region, and a fifth region, the fourth region provided between the second gate insulating film and the third region and in contact with the second gate insulating film, the fifth region provided between the third region and the fourth region, the third region containing titanium (Ti) and aluminum (Al), the fourth region containing the first metal element and nitrogen (N), and the fifth region containing nitrogen (N) and a second metal element being at least one of hafnium (Hf) and zirconium (Zr); and a memory cell array including a plurality of memory cells.
16. The semiconductor memory device according to claim 15, wherein the n-channel MOSFET and the p-channel MOSFET control operations of the plurality of memory cells.
17. The semiconductor memory device according to claim 15, wherein the first metal element and the second metal element are different.
18. The semiconductor memory device according to claim 15, wherein the first gate electrode further includes a sixth region provided between the first region and the second region and containing a third metal element and nitrogen (N), the third metal element being different from the first metal element and the second metal element, and the second gate electrode further includes a seventh region provided between the fourth region and the fifth region and containing the third metal element and nitrogen (N).
19. The semiconductor memory device according to claim 15, wherein the first metal element is a transition metal element.
20. The semiconductor memory device according to claim 19, wherein the first metal element is at

least one metal element selected from a group consisting of titanium (Ti), tantalum (Ta), vanadium (V), chromium (Cr), zirconium (Zr), niobium (Nb), hafnium (Hf), and tungsten (W).

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